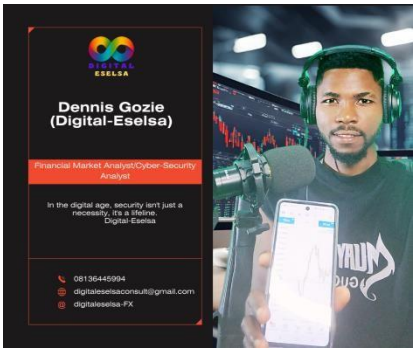
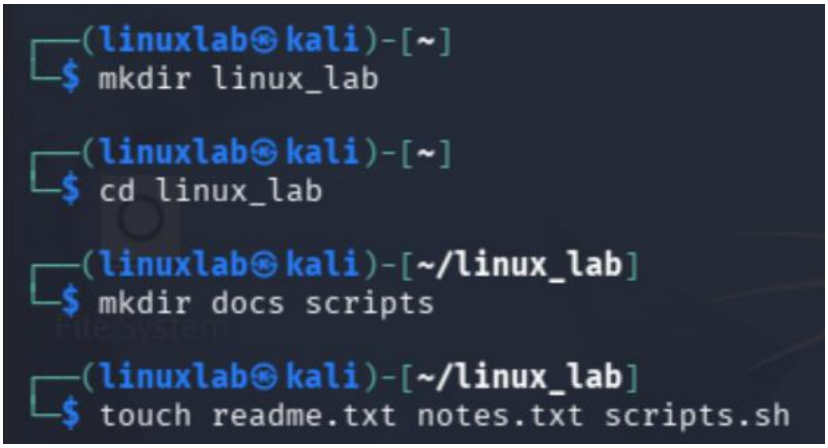
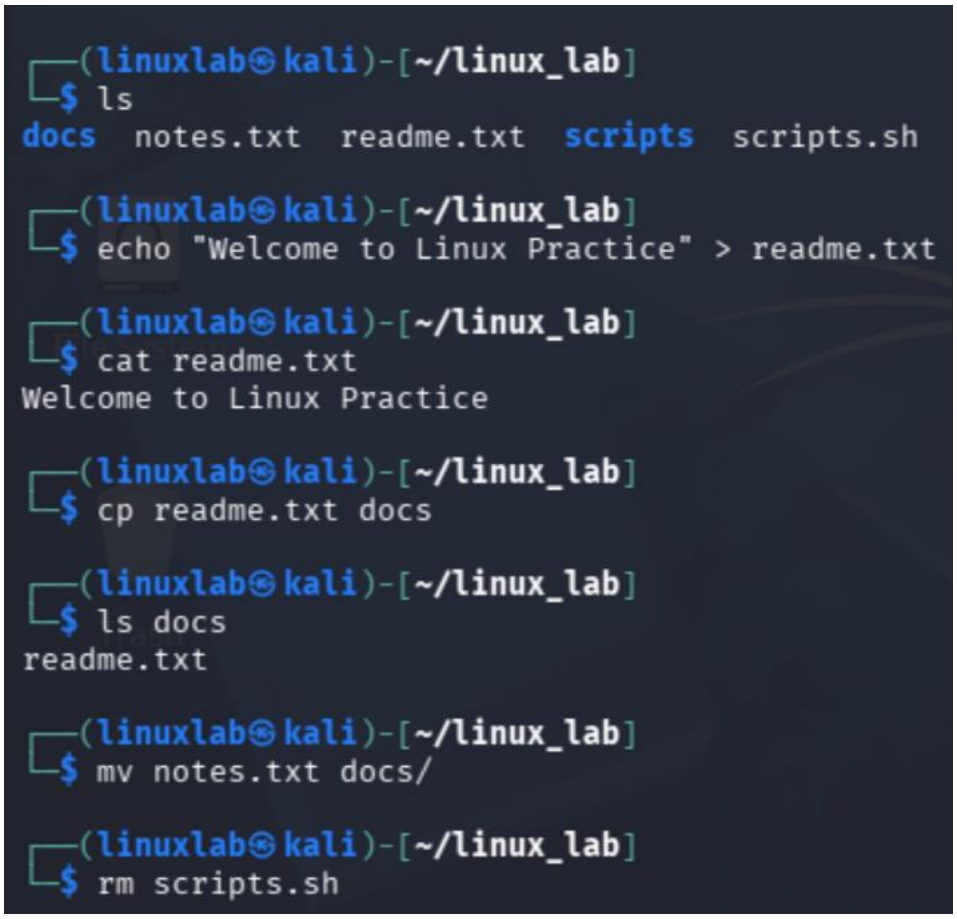
Linux Practical Test Package

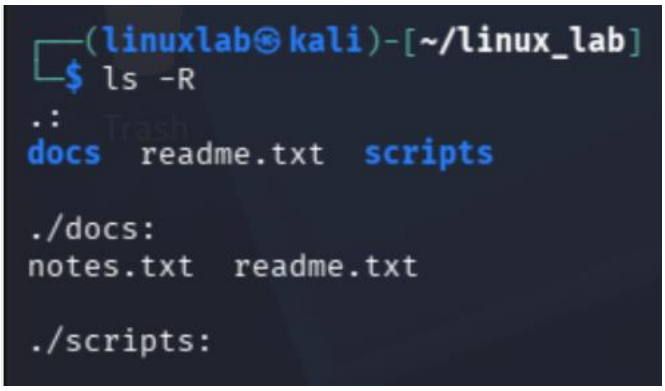


Activity 1: Navigation & File Management

- Create a directory named `linux_lab`.
- Inside it, create two subdirectories: `docs` and `scripts`.
- Create three files: `readme.txt`, `notes.txt`, `script.sh`. Write *"Welcome to Linux Practice"* inside `readme.txt`.
- Copy `readme.txt` to `docs`.
- Move `notes.txt` to `docs`. Delete `script.sh`.
- Display the directory structure with `tree` or `ls -R`.

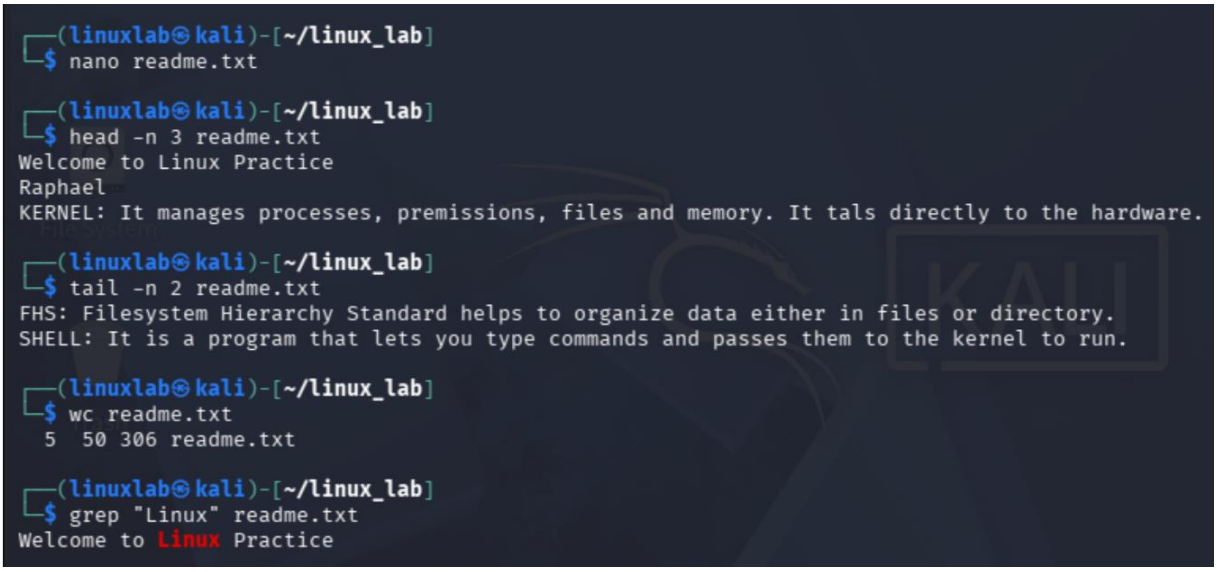






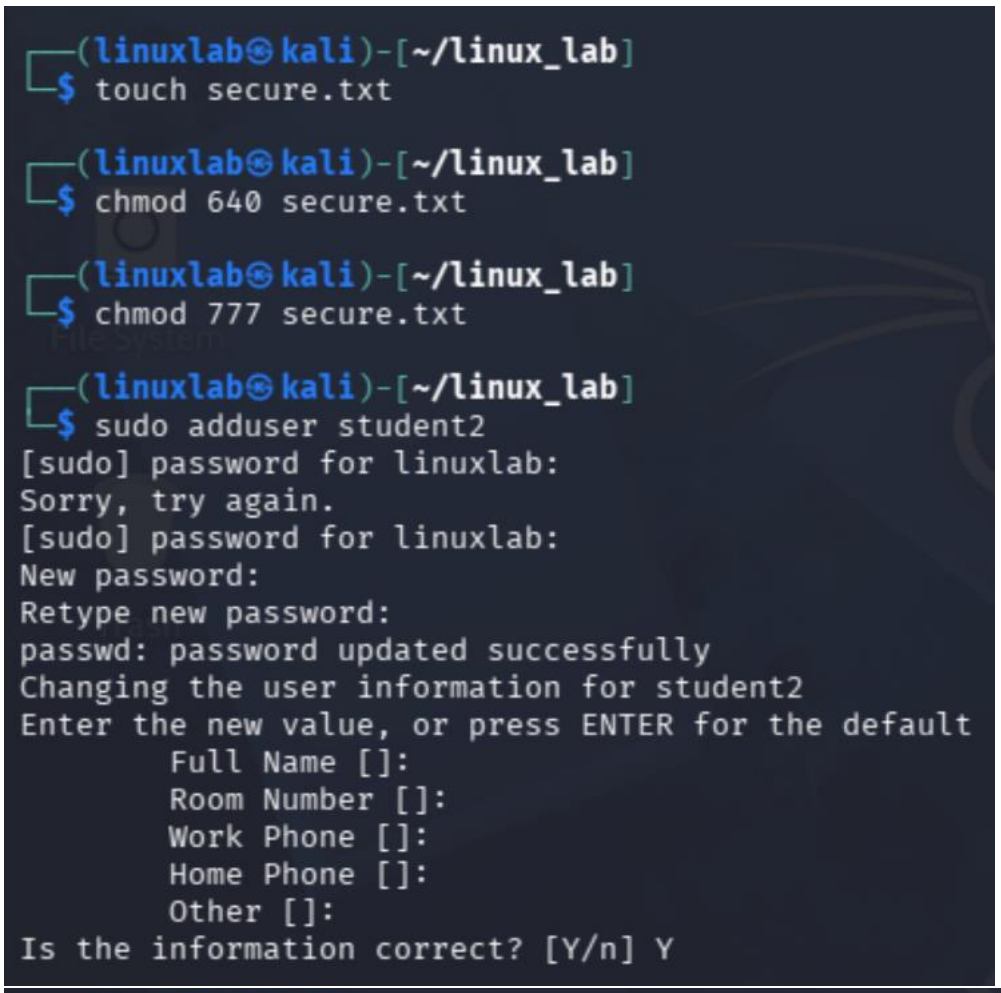
Activity 2: File Viewing & Editing

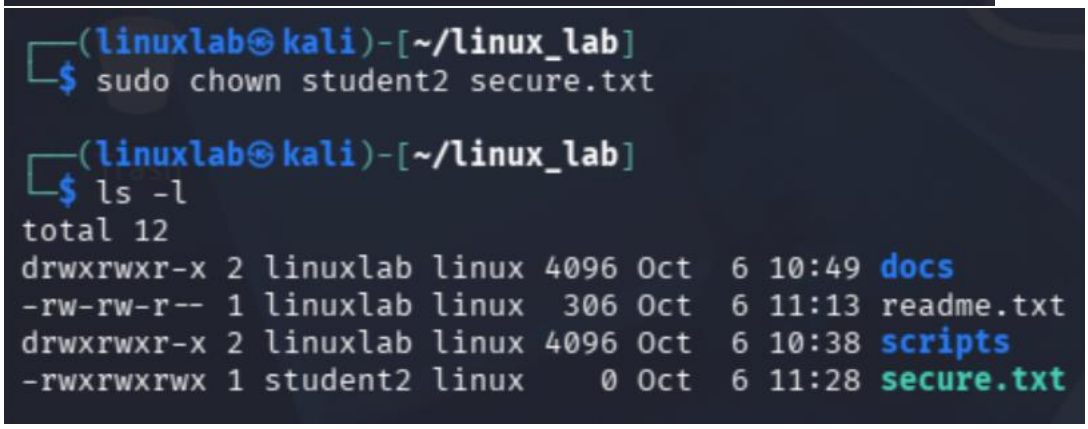
- Open `readme.txt` with `nano` and add your name at the bottom. Write the function of Kernel, FHS and Shell under.
- Display the first 3 lines of `readme.txt` with `head`. □ Display the last 2 lines with `tail`.
- Count how many words are in `readme.txt` using `wc`.
- Search for the word “Linux” inside `readme.txt` using `grep`.



Activity 3: Permissions & Ownership

- Create a file called `secure.txt`.
- Give the owner **read and write** permissions, group **read only**, others **no permissions**.
- Change permissions of `secure.txt` to **777**. Create a new user `student2` (if permitted). Change the owner of `secure.txt` to `student2`. Verify changes with `ls -l`.





Activity 4: Archiving & Compression

- Create a directory `backup_test` and place three text files in it. Create a tar archive called `backup.tar` from that directory.
- Compress `backup.tar` into `backup.tar.gz`.
- Extract `backup.tar.gz` into a folder named `restore_test`.
- Verify that all files are restored properly.

```
┌──(linuxlab㉿kali)-[~/linux_lab]
└─$ mkdir backup_test

┌──(linuxlab㉿kali)-[~/linux_lab]
└─$ cd ../backup_test
-bash: cd: ../backup_test: No such file or directory

┌──(linuxlab㉿kali)-[~/linux_lab]
└─$ cd backup_test

┌──(linuxlab㉿kali)-[~/linux_lab/backup_test]
└─$ touch one.txt two.txt three.txt

┌──(linuxlab㉿kali)-[~/linux_lab/backup_test]
└─$ tar -cvf backup.tar .
./
./one.txt
./two.txt
tar: ./backup.tar: archive cannot contain itself; not dumped
./three.txt

┌──(linuxlab㉿kali)-[~/linux_lab/backup_test]
└─$ gzip backup.tar
```

```
┌──(linuxlab㉿kali)-[~/linux_lab/backup_test]
└─$ mkdir ../restore_test

┌──(linuxlab㉿kali)-[~/linux_lab/backup_test]
└─$ tar -xzvf backup.tar.gz -C ../restore_test/
./
./one.txt
./two.txt
./three.txt

┌──(linuxlab㉿kali)-[~/linux_lab/backup_test]
└─$ ls restore_test
ls: cannot access 'restore_test': No such file or directory

┌──(linuxlab㉿kali)-[~/linux_lab/backup_test]
└─$ ls ../restore_test
one.txt  three.txt  two.txt
```

Activity 5: Process Management

- Run a background process using `sleep 300 &`.
- Use `jobs` to display background jobs.
- Bring the job to the foreground with `fg`. Kill the process using its PID.
- Use `ps aux | grep sleep` to confirm it is terminated.
- Use `top` to view running processes.

```
┌──(linuxlab㉿kali)-[~]
└─$ sleep 300 &
[1] 7531

┌──(linuxlab㉿kali)-[~]
└─$ jobs
[1]+  Running                 sleep 300 &

┌──(linuxlab㉿kali)-[~]
└─$ fg
-bash: fg: job has terminated
[1]+  Done                    sleep 300

┌──(linuxlab㉿kali)-[~]
└─$ kill 7531
-bash: kill: (7531) - No such process

┌──(linuxlab㉿kali)-[~]
└─$ ps aux | grep sleep
linuxlab   14639  0.0  0.1   6556  2388 pts/0    S+   04:56   0:00 grep --color=auto sleep
```

```
┌──(linuxlab㉿kali)-[~]
└─$ top

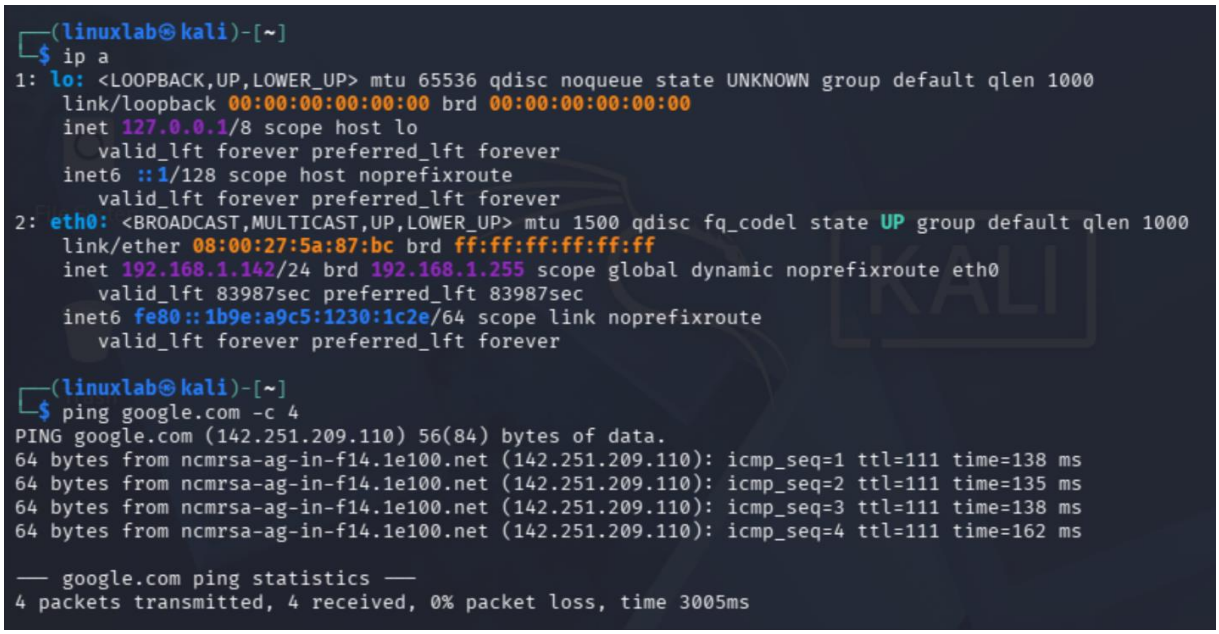
top - 05:03:51 up 33 min,  1 user,  load average: 0.16, 0.16, 0.17
Tasks: 177 total,   1 running, 176 sleeping,   0 stopped,   0 zombie
%Cpu(s):  1.7 us,  1.7 sy,  0.0 ni, 96.5 id,  0.0 wa,  0.0 hi,  0.2 si,  0.0 st
MiB Mem :   1969.3 total,    347.4 free,    788.8 used,   1008.9 buff/cache
MiB Swap:    953.7 total,    953.7 free,      0.0 used.   1180.6 avail Mem

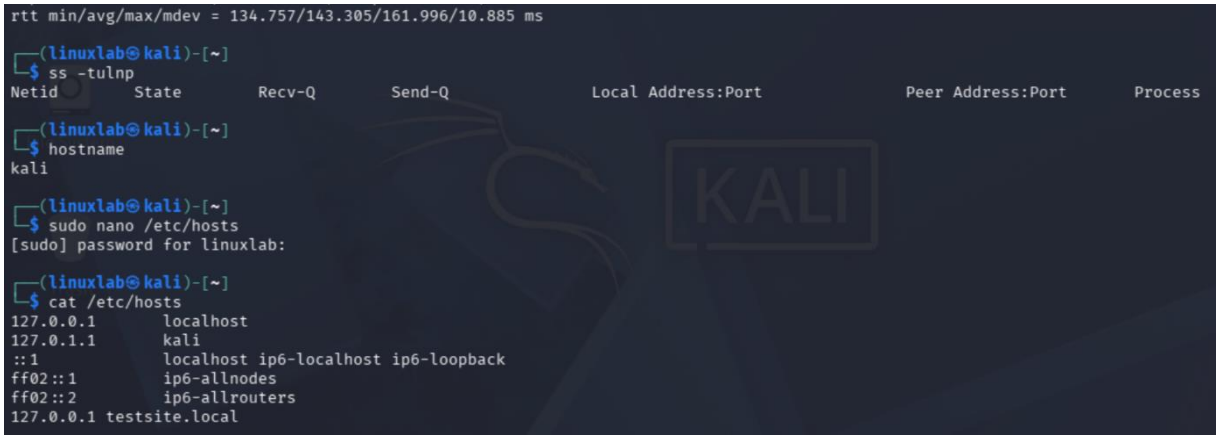
    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    651 root      20   0  417276 119476  76984 S   1.6   5.9   0:24.72 Xorg
   2376 kali      20   0  661236  68392  50852 S   1.0   3.4   0:02.84 qterminal
   1194 kali      20   0 1119740 139724  91212 S   0.7   6.9   0:07.61 xfwm4
   1303 kali      20   0  300948  49852  21940 S   0.7   2.5   0:11.50 wrapper-2.0
     14 root      20   0       0      0      0 S   0.3   0.0   0:00.19 ksoftirqd/0
    456 message+  20   0    9996   6664   4376 S   0.3   0.3   0:02.78 dbus-daemon
    571 root      20   0  291764   3596   3184 S   0.3   0.2   0:00.54 VBoxService
   1124 kali      20   0  215388   3336   2928 S   0.3   0.2   0:04.56 VBoxClient
   1305 kali      20   0  277080  30292  22964 S   0.3   1.5   0:07.72 wrapper-2.0
  16878 root      20   0       0      0      0 I   0.3   0.0   0:00.07 kworker/0:2-ata_sff
      1 root      20   0   25572  15608  11720 S   0.0   0.8   0:02.33 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.01 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
```

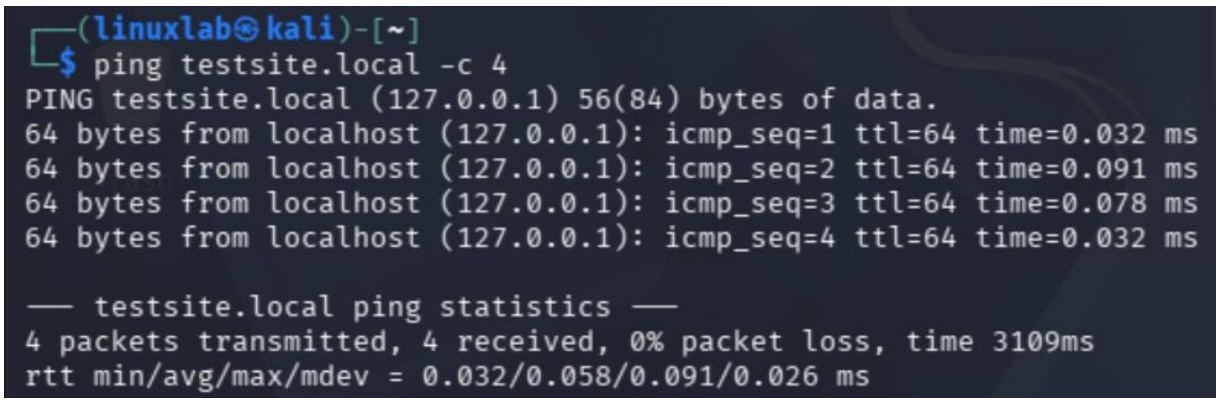
Activity 6: Networking Basics

- Display your system's IP address using `ip a`.

- Use `ping google.com -c 4` to test connectivity.
- Check which ports are listening using `ss -tulnp`. Find the hostname of your system with `hostname`.
- Add a new entry to `/etc/hosts` mapping `127.0.0.1 testsite.local`.
- Ping `testsite.local` to verify.

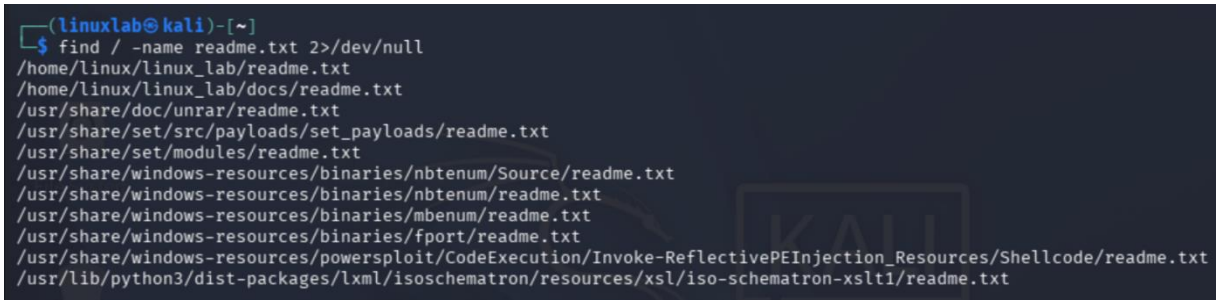


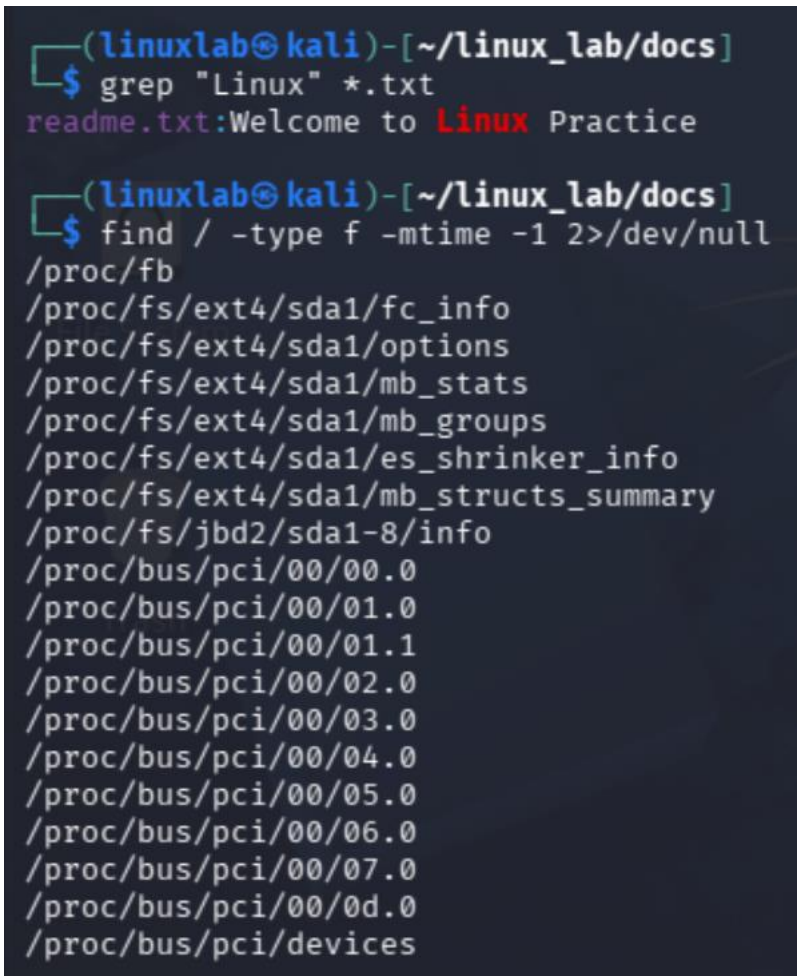




Activity 7: Searching & Finding

- Search for the file `readme.txt` using `find / -name readme.txt 2>/dev/null`.
- Search within all `.txt` files in `docs` for the word “Linux” using `grep`. Display all files modified in the last 24 hours using `find`.
- Count the number of `.txt` files in the home directory using `find` + `wc -l`.





```
/sys/module/drm/sections/__tracepoint_check
/sys/module/drm/sections/.orc_header
/sys/module/drm/sections/__ksymtab_gpl
/sys/module/drm/sections/___srcu_struct_ptrs
/sys/module/drm/sections/.altinstr_aux
/sys/module/drm/sections/__ex_table
/sys/module/drm/sections/__bpf_raw_tp_map
/sys/module/drm/sections/.ref.data
/tmp/.xfsm-ICE-M67KW3
/tmp/config-err-4klTsQ
/tmp/.X0-lock
/dev/shm/sem.haveged_sem

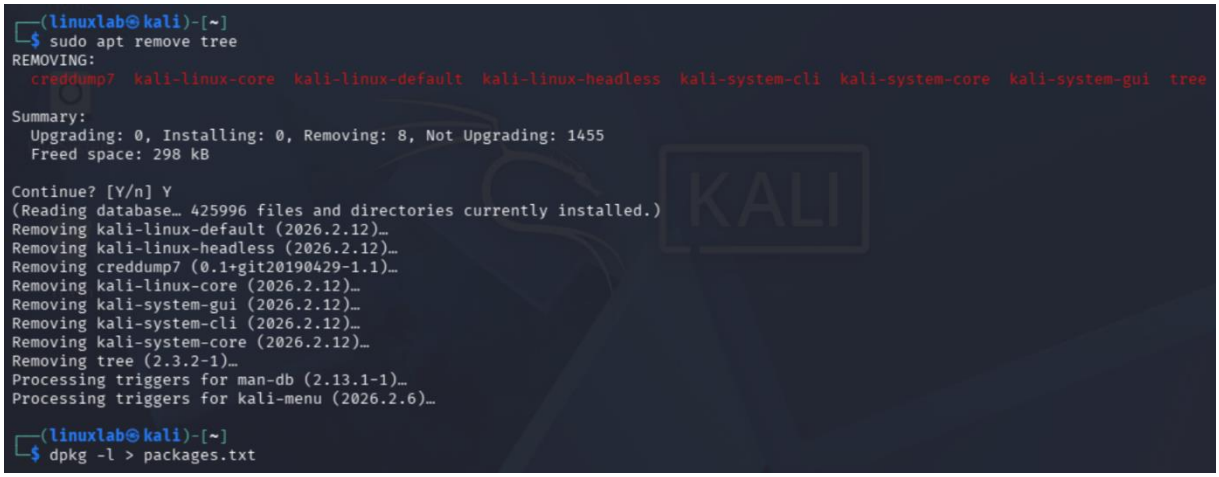
┌──(linuxlab㉿kali)-[~/linux_lab/docs]
└─$ find ~ -type f -name "*.txt" | wc -l
10
```

Activity 8: Package Management (Debian/Ubuntu-based)

- Update the system package list with `sudo apt update`.
- Install the package `tree`.
- Verify installation by running `tree --version`.
- Remove the `tree` package.
- List all installed packages and redirect the output to `packages.txt`.

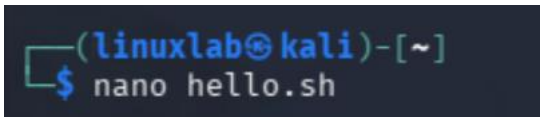
```
┌──(linuxlab㉿kali)-[~]
└─$ sudo apt update
[sudo] password for linuxlab:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.6 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [54.2 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [104 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [183 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [170 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [893 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [16.0 kB]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [42.3 kB]
Fetched 77.2 MB in 2min 19s (558 kB/s)
1461 packages can be upgraded. Run 'apt list --upgradable' to see them.

┌──(linuxlab㉿kali)-[~]
└─$ tree --version
tree v2.3.2 © 1996 - 2026 by Steve Baker, Thomas Moore, Francesc Rocher, Florian Sesser, Kyosuke Tokoro
```



Activity 9: Scripting Basics

- Create a script named `hello.sh`. □ Inside it, add commands to:
    1. Display “Hello, Linux World!”
    2. Show the current date and time (`date`) 3. Show the current working directory (`pwd`) □ Make the script executable.
- Run the script.





```
┌──(linuxlab㉿kali)-[~]
└─$ chmod +x hello.sh

┌──(linuxlab㉿kali)-[~]
└─$ ./hello.sh
Hello, Linux World!
Wed Oct  7 04:19:21 PM EDT 2026
/home/linux
```

Activity 10: System Monitoring & Logs

- Display disk usage using `df -h`.
- Check memory usage using `free -h`.
- Display the last 10 system log entries from `/var/log/syslog` (or `/var/log/messages`).
- Use `uptime` to show how long the system has been running.
- Show current logged-in users with `who`.
- Display your system's CPU information with `lscpu`.

```
┌──(linuxlab㉿kali)-[~]
└─$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            889M     0  889M   0% /dev
tmpfs           197M  968K  196M   1% /run
/dev/sda1        79G   16G   59G  22% /
tmpfs           985M  4.0K  985M   1% /dev/shm
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           985M  8.0K  985M   1% /tmp
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           197M  108K  197M   1% /run/user/1000

┌──(linuxlab㉿kali)-[~]
└─$ free -h
               total        used        free      shared  buff/cache   available
Mem:           1.9Gi       898Mi       290Mi       6.6Mi       971Mi       1.0Gi
Swap:          953Mi       9.9Mi       943Mi

┌──(linuxlab㉿kali)-[~]
└─$ tail -n 10 /var/log/syslog
tail: cannot open '/var/log/syslog' for reading: No such file or directory
```

```
┌──(linuxlab㉿kali)-[~]
└─$ sudo journalctl -n 10 --no-pager
Oct 07 16:37:51 kali sudo[66435]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1006)
Oct 07 16:39:01 kali CRON[66995]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Oct 07 16:39:01 kali CRON[66997]: (root) CMD (   [ -x /usr/lib/php/sessionclean ] && if [ ! -d /run/systemd/system ]; then /usr/lib/p
hp/sessionclean; fi)
Oct 07 16:39:01 kali CRON[66995]: pam_unix(cron:session): session closed for user root
Oct 07 16:39:30 kali systemd[1]: Starting phpsessionclean.service - Clean php session files...
Oct 07 16:39:30 kali systemd[1]: phpsessionclean.service: Deactivated successfully.
Oct 07 16:39:30 kali systemd[1]: Finished phpsessionclean.service - Clean php session files.
Oct 07 16:39:45 kali sudo[66435]: pam_unix(sudo:session): session closed for user root
Oct 07 16:41:05 kali sudo[68049]: linuxlab : TTY=pts/0 ; PWD=/home/linux ; USER=root ; COMMAND=/usr/bin/journalctl -n 10 --no-pager
Oct 07 16:41:05 kali sudo[68049]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1006)

┌──(linuxlab㉿kali)-[~]
└─$ uptime
 16:42:09 up  2:17,  1 user,  load average: 0.35, 0.30, 0.25

┌──(linuxlab㉿kali)-[~]
└─$ who
kali     seat0        2026-10-07 14:26 (:0)
```

```
┌──(linuxlab㉿kali)-[~]
└─$ lscpu
Architecture:                x86_64
  CPU op-mode(s):            32-bit, 64-bit
  Address sizes:             39 bits physical, 48 bits virtual
  Byte Order:                Little Endian
CPU(s):                      2
  On-line CPU(s) list:       0,1
Vendor ID:                   GenuineIntel
  Model name:                Intel(R) Core(TM) i5-6300U CPU @ 2.40GHz
    CPU family:              6
    Model:                   78
    Thread(s) per core:      1
    Core(s) per socket:      2
    Socket(s):               1
    Stepping:                3
    Microcode version:       0×cc
    BogoMIPS:                4992.00
    Flags:                   fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht sy
                             scall nx rdtscp lm constant_tsc rep_good nopl xtopology nonstop_tsc cpuid tsc_known_freq pni pclmulqdq
                             ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm
                             3dnowprefetch pti fsgsbase bmi1 avx2 bmi2 invpcid rdseed adx clflushopt arat md_clear flush_l1d
Virtualization features:
  Hypervisor vendor:         KVM
  Virtualization type:       full
Caches (sum of all):
```

This **10-activity lab series** ensures students cover:

- **Basics (files & navigation)**
- **Permissions & ownership**
- **Archiving & compression**
- **Processes & networking**
- **Searching & package management**
- **Scripting & system monitoring**

Raphael _ Linux Task04